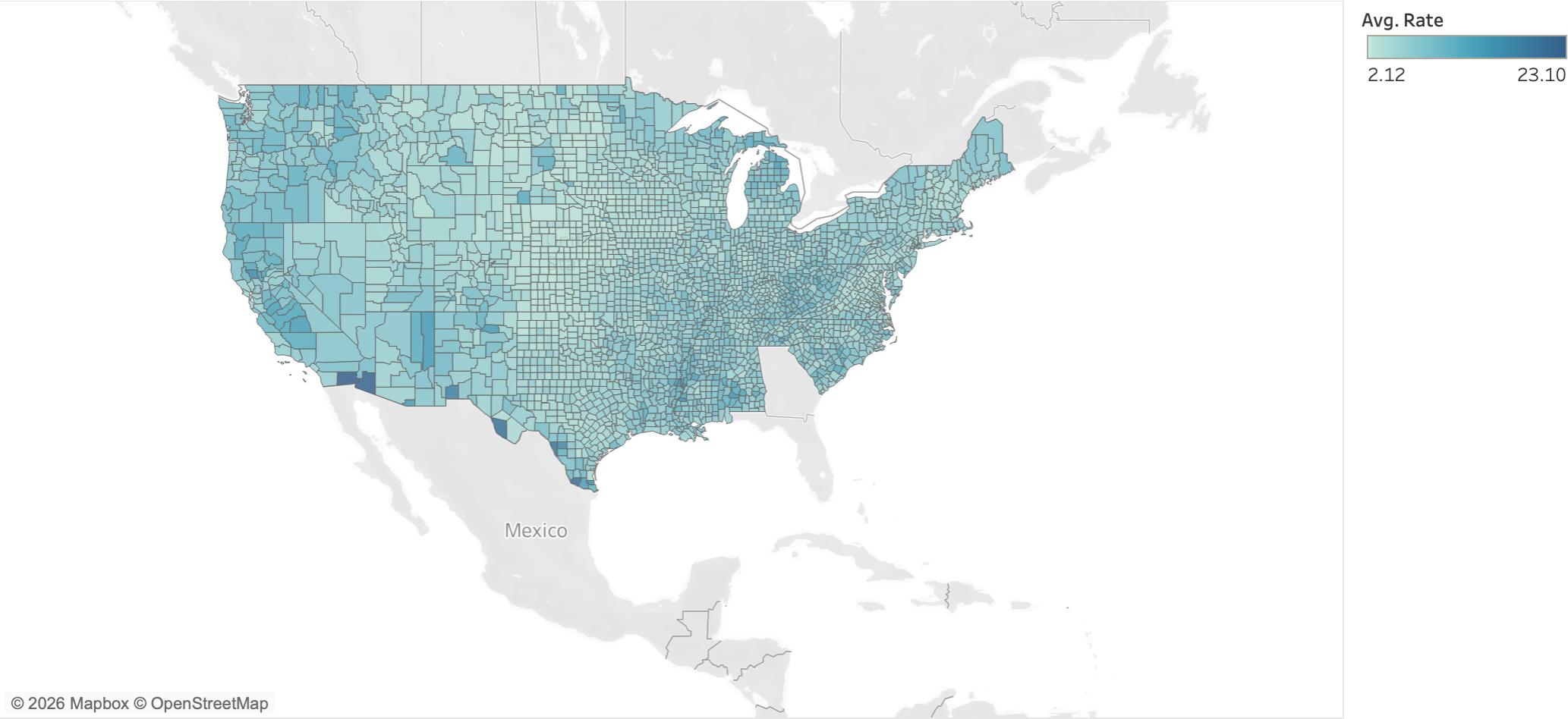


Module4

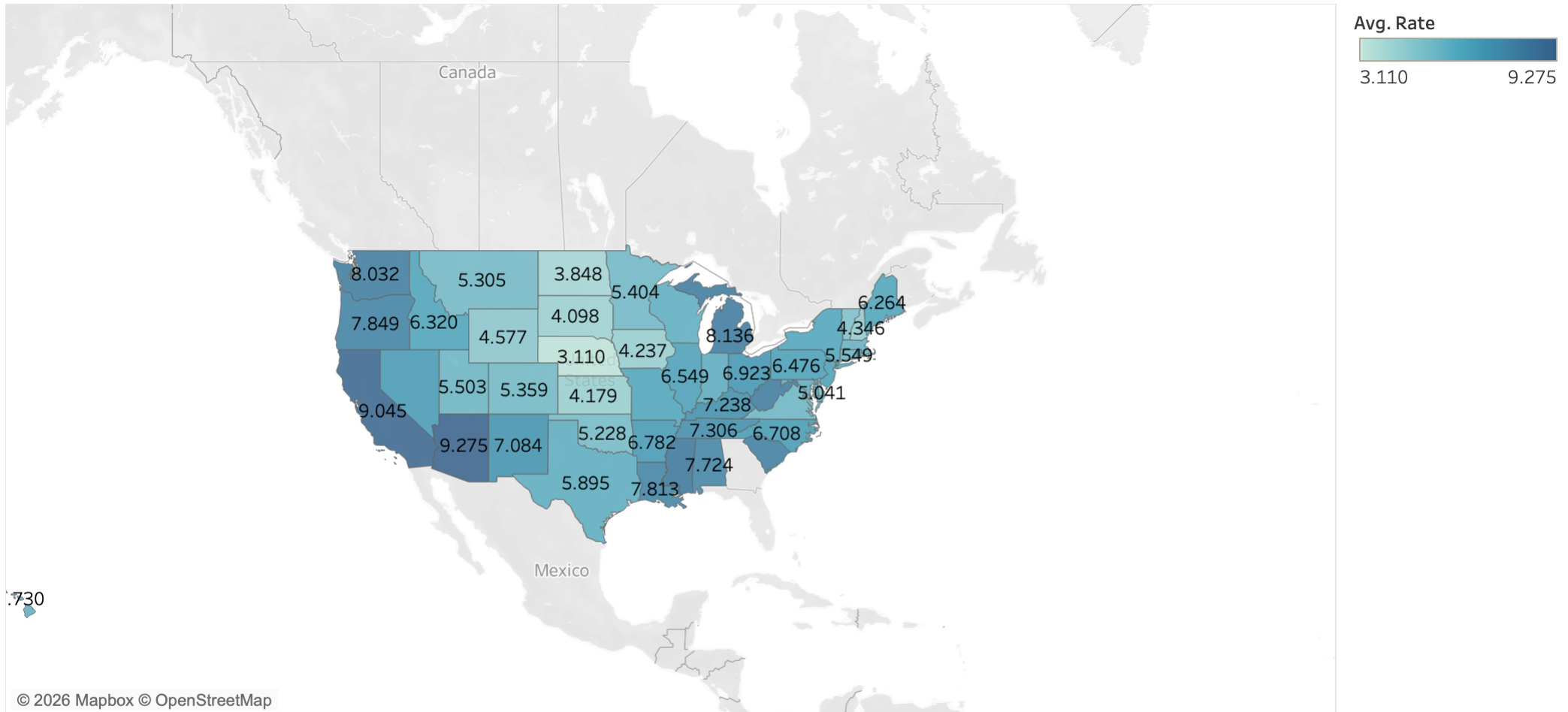
File created on: 7/27/26 1:29:56 PM PDT

US Geographic Unemployment by State County



Map based on Longitude (generated) and Latitude (generated). Color shows average of Rate. Details are shown for County and State. The data is filtered on Year, which ranges from 1990 to 2016.

US Geographic Unemployment by State

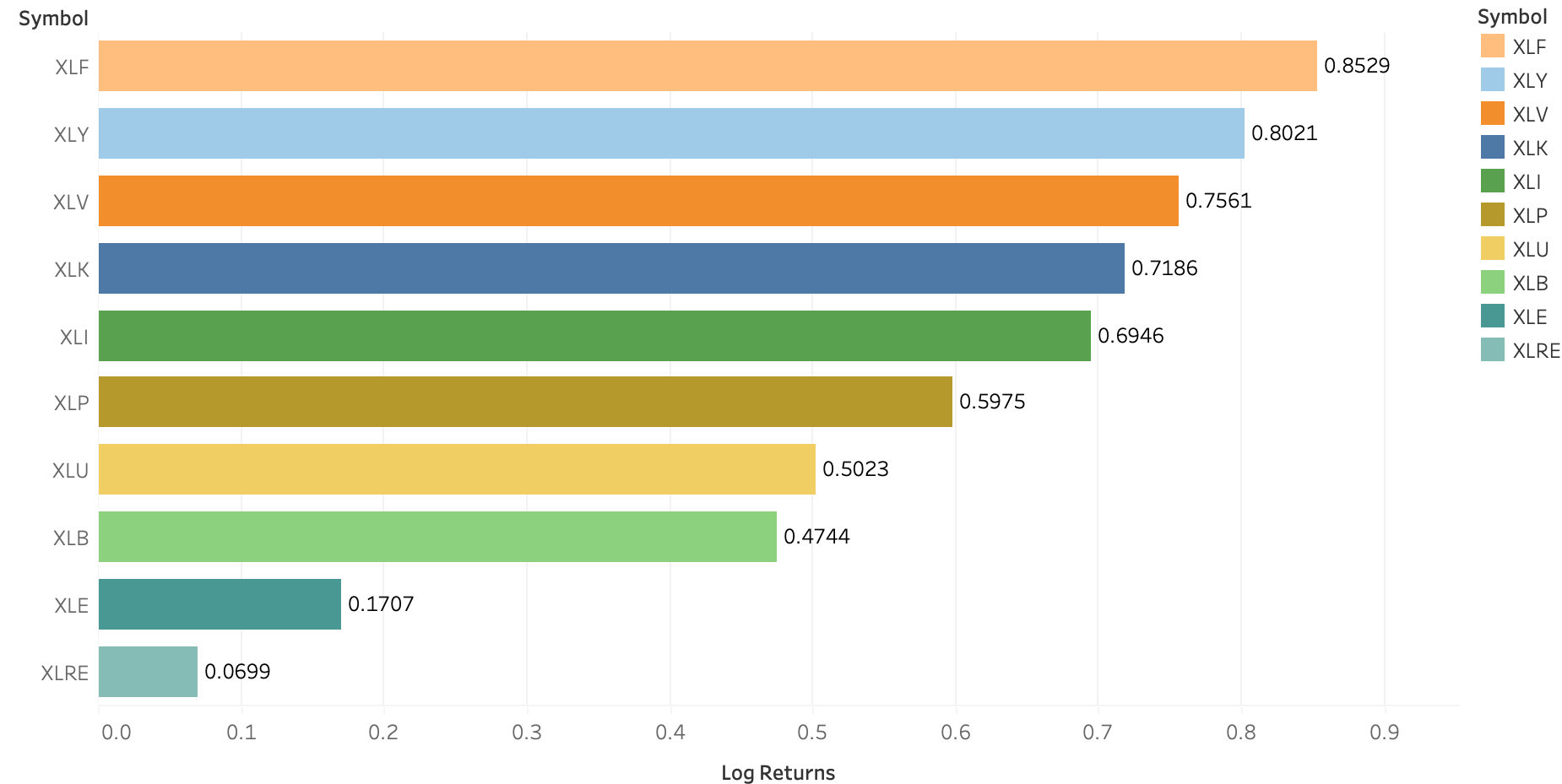


Question: Which states have the highest average unemployment rate?

Answer: The state with the highest unemployment rate is Arizona at 9.275%. California has the second highest rate at 9.045%.

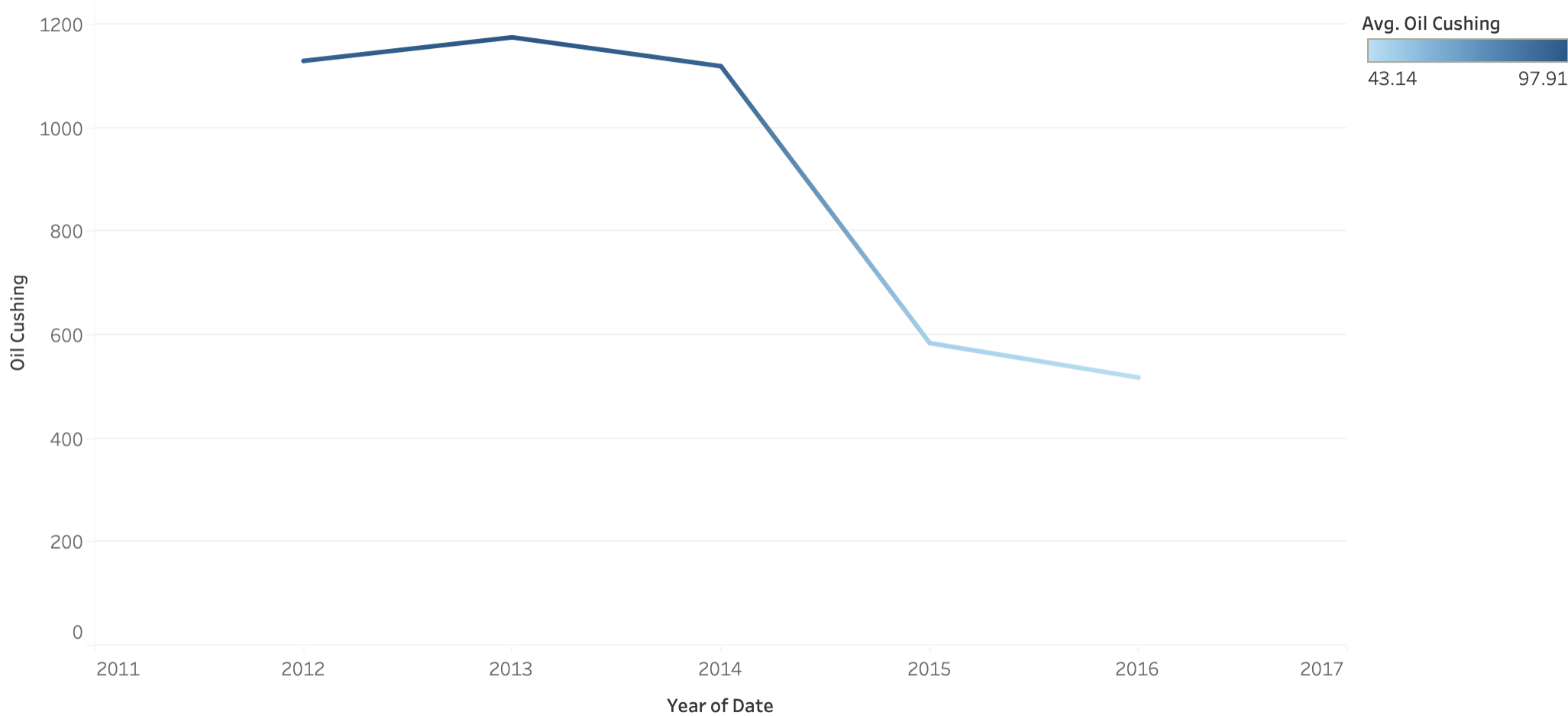
Map based on Longitude and Latitude. Color shows average of Rate. The marks are labeled by average of Rate. Details are shown for State. The data is filtered on Year, which ranges from 1990 to 2016.

Comparative Differences Return of Sector



Sum of Log Returns for each Symbol. Color shows details about Symbol. The marks are labeled by sum of Log Returns. The data is filtered on Date1 Year and Year. The Date1 Year filter ranges from 2012 to 2023. The Year filter ranges from 1990 to 2016. The view is filtered on Symbol, which keeps 11 members.

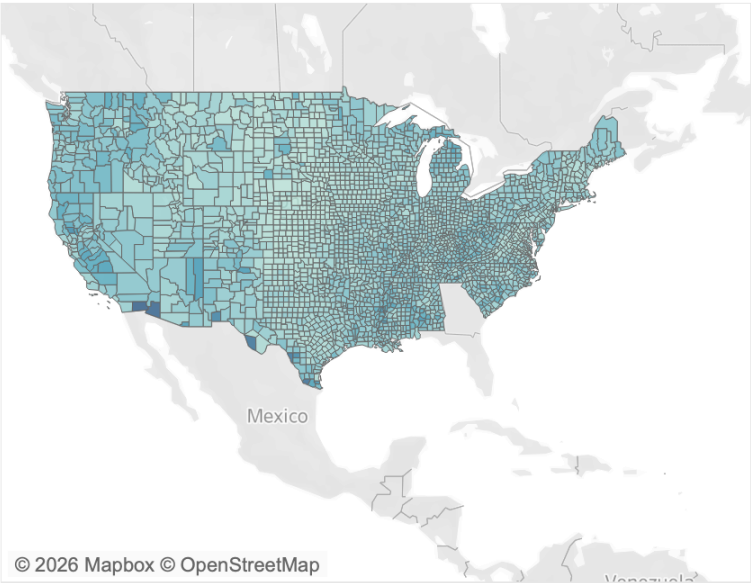
Question 5: Change in Oil Price by Year



The trend of sum of Oil Cushing for Date Year. Color shows average of Oil Cushing. The data is filtered on Year (oil!cushing) and Year. The Year (oil!cushing) filter ranges from 2012 to 2023. The Year filter ranges from 1990 to 2016.

Geospatial Analysis and Network Analysis

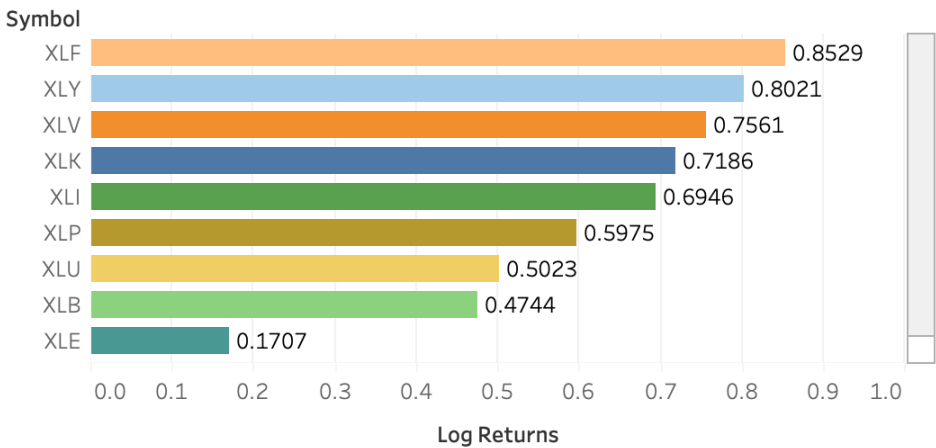
US Geographic Unemployment by State County



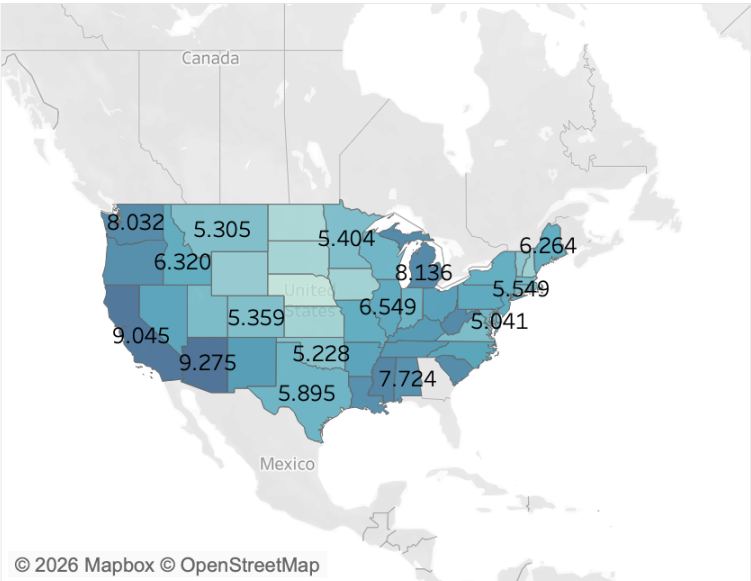
Avg. Rate



Comparative Differences Return of Sector



US Geographic Unemployment by State



Year Over Year Unemployment by State (Average 2007-2009)

State	Year							
	1990	1991	1992	1993	1994	1995	1996	1997
Alabama	8.23	9.08	9.30	9.18	7.62	7.61	7.24	6.9
Arizona	8.347	8.347	8.347	8.347	8.347	8.347	8.347	8.34
Arkansas	8.29	8.94	11.74	11.46	11.45	10.65	11.65	9.9
California	8.802	8.802	8.802	8.802	8.802	8.802	8.802	8.80
Colorado	7.64	8.30	7.97	6.96	6.18	5.67	6.43	6.3
Florida	6.974	6.974	6.974	6.974	6.974	6.974	6.974	6.97
Georgia	8.30	8.36	12.03	12.01	10.86	10.61	9.93	9.1
Idaho	9.291	9.291	9.291	9.291	9.291	9.291	9.291	9.29
Illinois	5.26	6.05	7.47	6.21	4.94	4.82	5.23	4.5

Average Recession



Year

1990 to 2016